Math Diary: Banach Spaces Can Be Motivated by Absolute Convergence

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When reading through Conway's [1] section on complex series, we find the following proposition:

Proposition 1. *Let $\sum a_n$ be a series in $\mathbb{C}$. If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.*

The proof relies solely on addition, the triangle inequality, and the completeness of $\mathbb{C}$. $\mathbb{C}$ and $\mathbb{R}$ are certainly not the only spaces in which such properties hold. Can we extend this proposition to a more general set of spaces? In particular, we would hope for a more general space to have the following properties:

1. An addition operation along with additive inverses
2. A norm which satisfies the triangle inequality
3. Cauchy completeness in the metric space defined by this norm

As it turns out, these properties almost identically match those of Banach spaces. For example, Rudin [2] defines a Banach space as a normed linear space which is complete in the metric space defined by its norm. This is exactly what we were hoping for from such a space!

The proof from Conway [1] carries through with nearly no modification to the general case.

Proposition 2. *Let $\sum a_n$ be a series in a Banach space. If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.*

Proof. Let $\epsilon > 0$ and put $x_n = x_0 + x_1 + ... + x_n$. Since $\sum \|a_n\|$ converges there is an integer N such that $\sum_{n=N}^{\infty} \|a_n\| < \epsilon$. Thus, if $m > k \geq N$, we have (by the triangle property in the Banach space, not just $\mathbb{C}$!) that

$$\|x_m - x_k\| = \| \sum_{n=k+1}^{m} a_n \| \leq \sum_{n=k+1}^{m} \|a_n\| < \sum_{n=N}^{\infty} \|a_n\| < \epsilon. \tag{1}$$

The sequence $\{x_n\}$ is therefore a Cauchy sequence. Since our Banach space is complete, this sequence converges. Thus, the series converges in our Banach space, as desired. □

Actually, we could generalize this even further than Banach spaces. The definition of a linear space involves multiplication by a scalar; our proof involves no such multiplication.

References

[1] John B. Conway, *Functions of One Complex Variable I*, 2nd ed., Graduate Texts in Mathematics, vol. 11, Springer-Verlag, New York, 1978

[2] Walter Rudin, *Real and Complex Analysis*, 3rd ed., McGraw-Hill, New York, 1987.